An effect you can only see sideways

Does a training programme raise earnings, when the people who enrol are exactly the people who would have earned more anyway?

Abstract

Objective. Does a training programme raise earnings, when the people who enrol are exactly the people who would have earned more anyway? **Methods.** Write down the world, including the part nobody measured. 5 step(s) are recorded in full below. **Results.** OLS overstates the programme by +0.35, and the chart of sample sizes shows that no amount of data would have helped: the naive line converges on 2.32 against a truth of 2.00. Bias is not a small-sample problem. **Conclusions.** No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: Does a training programme raise earnings, when the people who enrol are exactly the people who would have earned more anyway?

2. Methods

Question. Does a training programme raise earnings, when the people who enrol are exactly the people who would have earned more anyway?

Write down the world, including the part nobody measured

The obstacle here is not a missing column, it is a structure: drive and circumstance push both enrolment and earnings, and they were never recorded. Writing that down as an edge — X <-> Y, a latent common cause — is what makes the problem statable. An analysis that cannot say what is confounding it cannot say what would fix it either.

Considered instead. Starting from the estimator. 'We ran 2SLS with distance as an instrument' skips the only question that determines whether the number means anything: does the graph you believe in actually license that move?

```
graph : X -> Y, Z -> X, X <-> Y
measured : ['X', 'Y', 'Z']
X <-> Y is the bidirected edge: a common cause of enrolment and earnings
that is not in the data and never will be. Z is the shifter -- distance
to
the nearest centre, say -- which moves enrolment for reasons of
geography.
```

Ask whether the effect is recoverable at all, before estimating anything

identify() is a question about the graph, not about the data, and it can be asked before a single row is collected. It returns the route, the instrument it found, and — the part that matters most in a referee report — the assumptions the route rests on, each tagged with whether the data can check it.

Considered instead. Nominating the instrument yourself and checking the first-stage F. That tests relevance, which is the assumption you can test, and quietly skips exclusion, which is the one that actually fails in practice.

```
status : downgraded
route : instrument
instrument : Z (found in the graph, not nominated by hand)
```

assumption	state	what it claims	what would challenge it
exclusion	satisfied	Z affects Y only through X	a, , d, i, r, e, c, t, , o, r, , c, o, n, f, o, u, n, d, e, d, , p, a, t, h, , i, n, , t, h, e, , g, r, a, p, h
relevance	unverified	Z is associated with X (first-stage strength)	f, i, r, s, t, -, s, t, a, g, e, , F, , b, e, l, o, w, , 1, 0
effect_homogeneity_or_monotonicity	unverified	the IV estimand equals the average effect (homogeneous/linear effect) or is read as a LATE under monotonicity	h, e, t, e, r, o, g, e, n, e, o, u, s, , e, f, f, e, c, t, s, ;, , d, e, f, i, e, r, s

Table 1. The verdict carries its own assumptions. 'unverified' is not a weasel word — it is the correct state for a claim no dataset can settle

Estimate it twice: once naively, once through the instrument

Both estimators are run on the same 5,000 workers so the difference between them is the method and nothing else. The naive comparison is not a straw man — it is what a well-run analysis of this dataset would report if nobody had drawn the graph.

Considered instead. Reporting only the instrumented estimate. The gap between the two is the size of the selection, and it is the most informative number on the page: it says how much of the raw programme effect was never the programme.

```
naive OLS 2.347 +/- 0.011 [ 2.326, 2.367] error +0.347
2SLS 1.999 +/- 0.021 [ 1.958, 2.040] error -0.001
truth 2.000
```

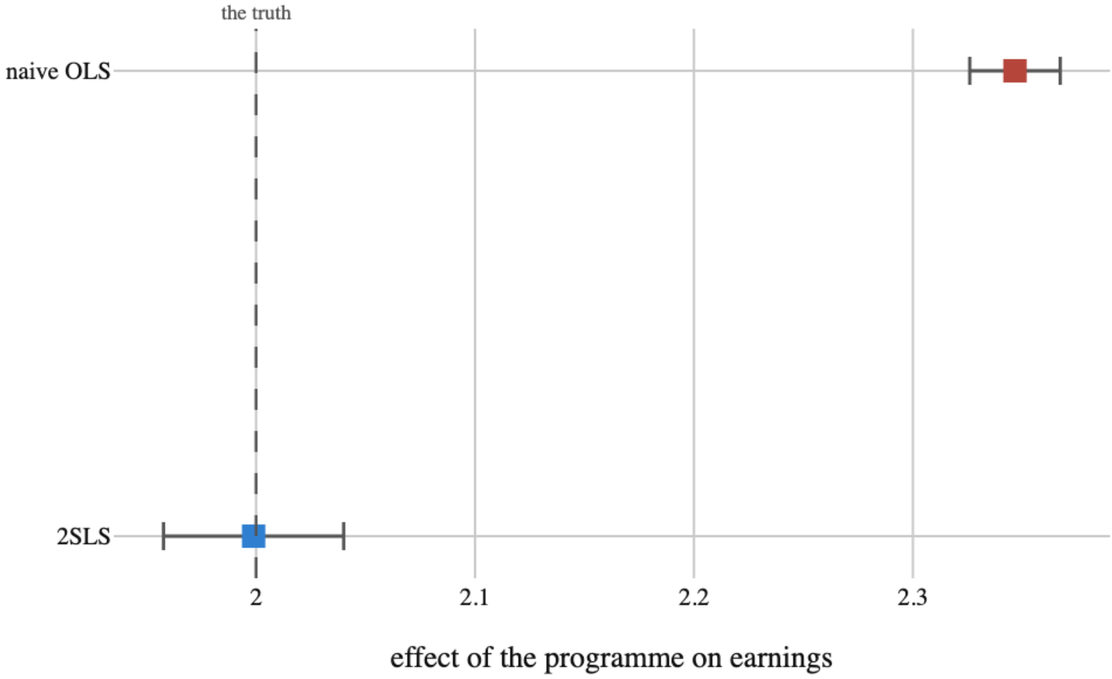

Figure 1. What each method says the programme is worth. The naive interval does not merely miss the truth, it excludes it confidently — the interval is narrow and in the wrong place. 2SLS lands on the truth and pays for it in width

Watch what more data does to each one

This is the distinction the whole example turns on. Noise shrinks as the square root of n; bias does not shrink at all. Re-running both estimators across sample sizes from 250 to 32,000 draws the difference rather than asserting it — the naive line converges, confidently, on the wrong number.

Considered instead. Saying 'OLS is biased' and moving on. Every analyst agrees with that sentence in the abstract and then, under deadline, argues that a large enough sample makes it academic. The chart is the answer to that argument.

workers	naive OLS	2SLS
250	2.292	2.117
500	2.291	2.046
1,000	2.318	2.014
2,000	2.316	1.961

4,000	2.343	1.985
8,000	2.334	2.035
16,000	2.331	2.001
32,000	2.319	1.980

Table 2

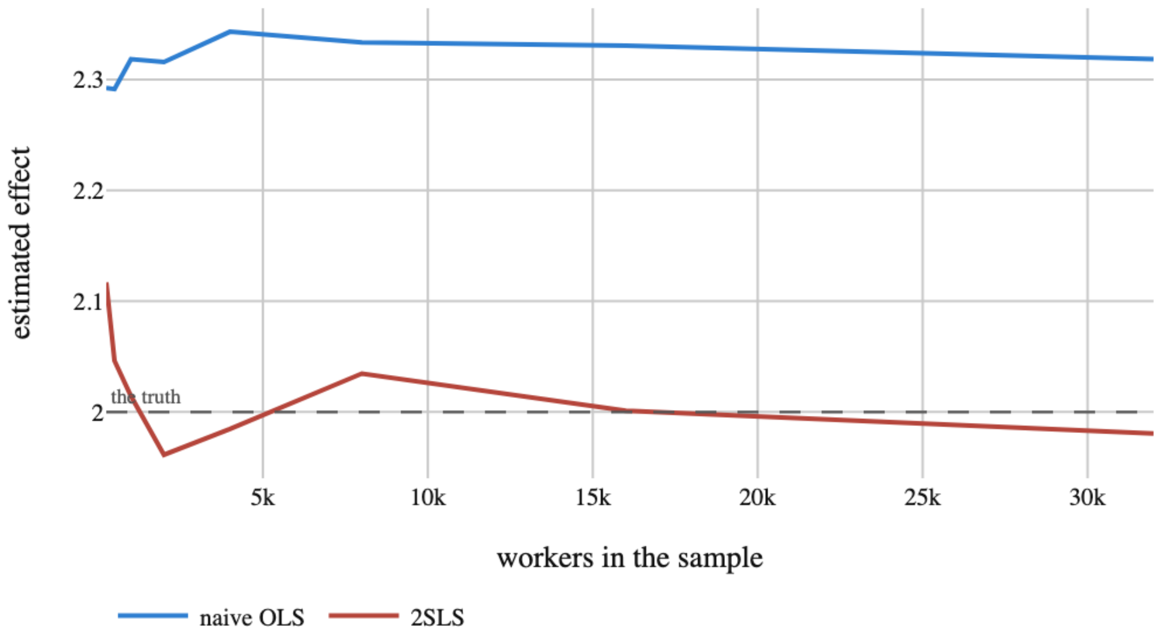

Figure 2. More data does not fix a bias. Both lines settle down. Only one settles on the right answer. Everything the naive estimator gains from a larger sample it spends on being more certain of the selection it cannot see

Check the instrument is actually doing work

A weak instrument is not a neutral loss of precision. When the first stage barely moves, 2SLS becomes biased towards OLS and its intervals stop covering, so a weak-instrument analysis can be worse than the naive one it was meant to replace. The first-stage F is the standard diagnostic, and axiom returns it as an assumption with a state rather than a number to eyeball.

Considered instead. Assuming relevance because the instrument is clever. Cleverness is an argument for exclusion, which cannot be tested; relevance is the part that can be, so it should be.

```
first-stage F : 2214.6
relevance : instrument_strength -> satisfied
a nearly irrelevant instrument (Z -> X of 0.05 instead of 1.0):
first-stage F : 0.1
relevance : violated
estimate : 1.810 +/- 3.968
```

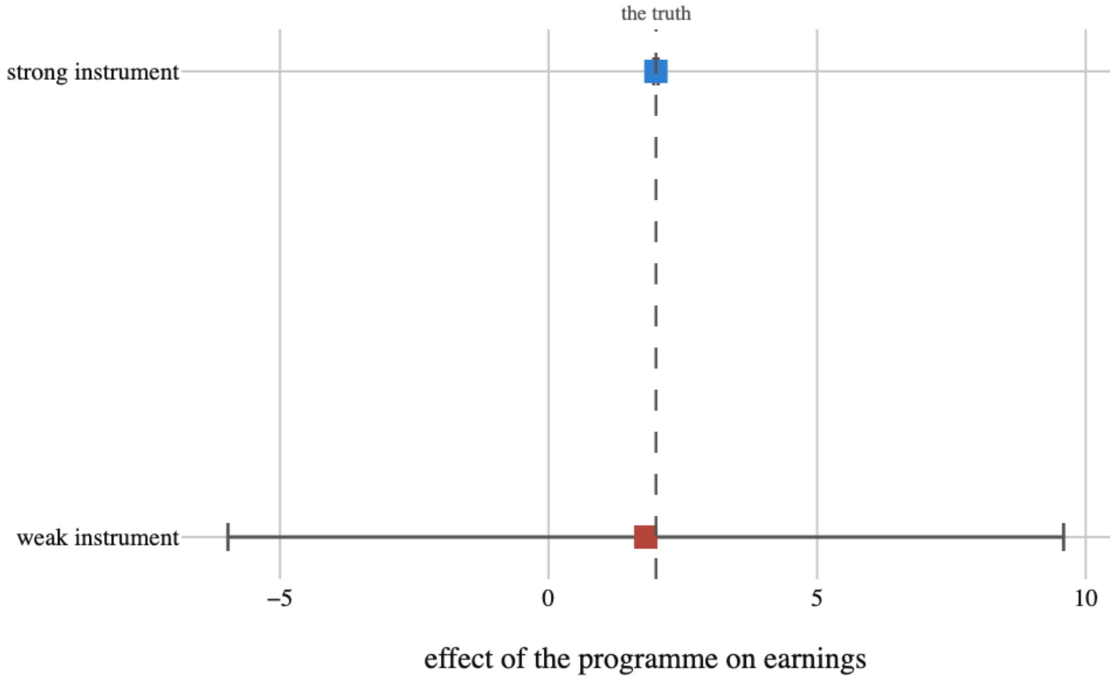

Figure 3. The same estimator, on an instrument that barely moves enrolment. Same graph, same route, same code. Only the strength of the first stage changed, and the answer became useless — which is why relevance is checked rather than assumed

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

OLS overstates the programme by +0.35, and the chart of sample sizes shows that no amount of data would have helped: the naive line converges on 2.32 against a truth of 2.00. Bias is not a small-sample problem.

What 2SLS bought was accuracy, and what it cost was width: its interval is 2.0 times wider than the naive one. That is the honest price of using only the part of the variation in enrolment that geography explains, and it is the right trade — a wide interval around the truth is usable, a narrow one around the wrong number is not.

The assumption table is the part to take to a referee. Relevance was checked and passed. Exclusion — that distance to a training centre affects earnings only through enrolment — is recorded as unverified, because it is unverifiable from this data. If it fails, everything above fails with it, and the verdict says so rather than burying it in a methods appendix.

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

Does a training programme raise earnings, when the people who enrol are exactly the people who would have earned more anyway?

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.